

Appendix G



Department of Computer Science & Engineering

Academic Year 2025-26

IV B. Tech I Semester Project work

B. Tech Project Proposal

Name of the Supervisor: Dr. M. Parthiban, Professor

I. Project Group Details

S.NO	Batch Code	Register No	Name of the Student
1	22CSEA16	22K61A05A9	Mutyala Hari Priyanka
2	22CSEA16	22K61A0512	Batthula Pavan Teja
3	22CSEA16	22K61A05G2	Thommandru Ratna Kumari
4	22CSEA16	22K61A05B4	Narim Subbalakshmi

Tentative Title: Cloud-Enabled AI Framework for Optimizing the Software Development Lifecycle

Domain / Sub-Domain: Artificial Intelligence / Software Engineering

Area of the Project: AI-powered SDLC optimization using cloud computing for intelligent software project management.

II. Brief Description of the Project

1. Problem Statement

Managing the System Development Lifecycle (SDLC) involves coordinating diverse activities, stakeholders, and resources. This complexity often leads to human errors, inefficient resource usage, project delays, and increased business costs. Traditional approaches heavily rely on manual intervention and lack the scalability and adaptability needed to manage modern SDLC tasks effectively.

2. Objective

To design and implement an AI-assisted framework leveraging transformer-based Large Language Models (LLMs) that support SDLC management by reducing manual effort, improving decision-making, optimizing resource allocation, and significantly minimizing overall business costs.

3. Scope of the Project

Covered:

- Automation of SDLC phase analysis and reporting.
- Project risk and priority classification.
- Resource optimization and task scheduling.
- Integration of transformer-based models trained with SDLC documentation.

- Performance evaluation through confusion matrix, perplexity, and cost-saving metrics.

Excluded:

- Real-time learning/adaptability of the model (currently static).
- Handling of non-English or multilingual project documents.
- In-depth interpretability or explainable AI (XAI) modules.
- Cloud deployment architecture (though adaptable for cloud).

4. Proposed Methodology of Solution

- Data Preparation: Extract and structure SDLC-related content from large documentation into a supervised dataset with context-response pairs.
- Model Design: Fine-tune a GPT-J-based transformer with over 150M parameters using transfer learning.
- Implementation: Develop three functional modules: user input interface, data preprocessing, and transformer-based AI-Analyst for output generation.
- Evaluation: Employ metrics like accuracy, precision, recall, F1-score, and perplexity; validate using confusion matrices and k-fold cross-validation.
- Cost Analysis: Simulate real-world SDLC projects to compare traditional vs AI-optimized workflows.

5. Tools / Technologies to be Used

- Programming: Python
- Libraries & Frameworks: PyTorch, Hugging Face Transformers, NumPy, Pandas, Scikit-learn
- Hardware: NVIDIA RTX A6000 GPU, Intel Xeon W-2295 CPU, 256 GB RAM
- OS: Ubuntu 20.04 LTS
- Pretrained Model: GPT-J (remodeled via transfer learning)
- Data Extraction: PyPDF2
- Evaluation Tools: Confusion Matrix, Perplexity analysis, K-fold validation

6. Expected Program Outcome

- SDLC phase and priority classification accuracy above 91%.
- Over 50% business cost savings demonstrated in replicated experiments.
- Reduced manual labor and rework through intelligent task recommendations.
- Efficient resource scheduling and project risk mitigation.
- Ready-to-integrate modular system adaptable for cloud deployment.

7. Significance / Societal Relevance

- Business Impact: Significantly reduces SDLC management costs through intelligent automation.
- Operational Efficiency: Streamlines project planning and execution, reducing errors and delays.
- Sustainability: Aligns with ESG goals by reducing resource wastage and Power usage.
- Inclusivity: Enables less experienced project managers to execute tasks effectively using AI insights.
- Innovation: Pioneers the integration of LLMs in full-lifecycle SDLC analysis.

III. Time Line for the Period (Semester week number)

Activity	Period
Title Confirmation	W1
Literature Survey	W2-3
Gap Identification	W4
Problem Formulation	W5
Survey Paper Publication	W6-7
Design	W8-9
Methodology	W10
Implementation	W11-13
Results	W14
Documentation	W15-16
Publication	W17-18

IV. Project Classification

Type of the Project: Application + Research

V. Quality of the Project

S.No	Quality Parameter	Description
1	Environment Safety	Serverless deployment minimizes energy consumption.
2	Ethics	Transparent and bias-aware AI integration; no data misuse.
3	Cost Effective	Open-source tools and freemium cloud services used.
4	Application	High relevance in industry, academia, and tech entrepreneurship.

VI. Project (Title) – Program Outcomes (POs) Relationship Matrix

Project Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
1.Design skills for engineering problems	X	X	X	X	X	X	X	X	X	X	X	X
2.Literature survey on current research areas	X	X	X	X	X	X	X	X	X	X	X	X
3.Apply software knowledge to meet industry needs	X	X	X	X	X	X	X	X	X	X	X	X
4.Research on innovative ideas for lifelong learning	X	X	X	X	X	X	X	X	X	X	X	X
5.Teamwork & communication skills	X	X	X	X	X	X	X	X	X	X	X	X
6.Platform for employability	X	X	X	X	X	X	X	X	X	X	X	X

VII. PEOs–Course Outcomes Relationship Matrix

Project Outcomes	PSO1	PSO2
1. Design skills for real-world problems	X	X
2. Research through literature survey	X	X
3. Software/hardware integration for industry standards	X	X
4. Interest in research and lifelong learning	X	X
5. Team and communication skills	X	X
6. Employability platform development	X	X

Signature of the Supervisor

DPRC Comment:

To be reviewed and approved by DPRC Committee

Signatures of DPRC Member 1

DPRC Member 2

DPRC Member 3

Signature of the HOD

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Date: _____